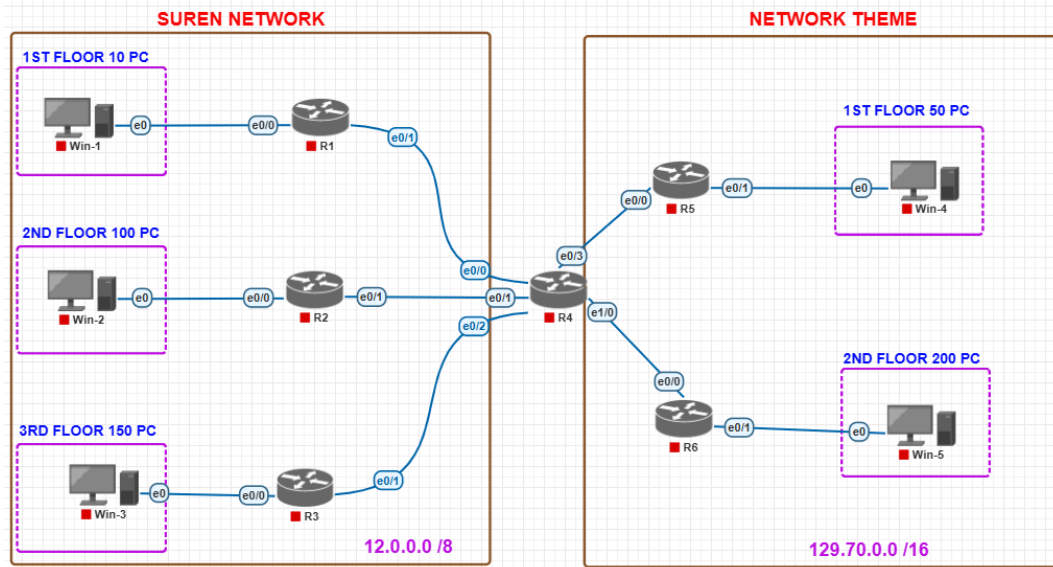




VLSM Subnetting Scheme — SUREN NETWORK Base Network 12.0.0.0/8							
Department/Link	Hosts Needed	Subnet Mask	CIDR	Network ID	First Usable IP	Last Usable IP	Broadcast IP
3rd Floor (150 Hosts)	150	255.255.255.0	/24	12.0.0.0	12.0.0.1	12.0.0.254	12.0.0.255
2nd Floor (100 Hosts)	100	255.255.255.128	/25	12.0.1.0	12.0.1.1	12.0.1.126	12.0.1.127
1st Floor (10 Hosts)	10	255.255.255.240	/28	12.0.1.128	12.0.1.129	12.0.1.142	12.0.1.143
Router1 - Router4 Link	2	255.255.255.252	/30	12.0.1.144	12.0.1.145	12.0.1.146	12.0.1.147
Router2 - Router4 Link	2	255.255.255.252	/30	12.0.1.148	12.0.1.149	12.0.1.150	12.0.1.151
Router3 - Router4 Link	2	255.255.255.252	/30	12.0.1.152	12.0.1.153	12.0.1.154	12.0.1.155
Next Available Network ID: 12.0.1.156 (remaining address space for future use)							

VLSM Subnetting Scheme — NETWORK THEME Base Network 129.70.0.0/16							
Department/Link	Hosts Needed	Subnet Mask	CIDR	Network ID	First Usable IP	Last Usable IP	Broadcast IP
2nd Floor (200 Hosts)	200	255.255.255.0	/24	129.70.0.0	129.70.0.1	129.70.0.254	129.70.0.255
1st Floor (50 Hosts)	50	255.255.255.192	/26	129.70.1.0	129.70.1.1	129.70.1.62	129.70.1.63
Router4 - Router5 Link	2	255.255.255.252	/30	129.70.1.64	129.70.1.65	129.70.1.66	129.70.1.67
Router4 - Router6 Link	2	255.255.255.252	/30	129.70.1.68	129.70.1.69	129.70.1.70	129.70.1.71
Next Available Network ID: 129.70.1.72 (remaining address space for future use)							

SUREN NETWORK (12.0.0.0/8)

3rd Floor → 150 pc (ip)

Network ID = 12.0.0.0/24

F. Ip = 12.0.0.1/24

L. Ip = 12.0.0.254/24

B. Ip = 12.0.0.255/24

2nd Floor → 100 - pc (ip)

Network ID = 12.0.1.0/25

F. Ip = 12.0.1.1/25

L. Ip = 12.0.1.126/25

B. Ip = 12.0.1.127/25

1st Floor → 10 pc (ip)

Network ID = 12.0.1.128/28

F. Ip = 12.0.1.129/28

L. Ip = 12.0.1.142/28

B. Ip = 12.0.1.143/28

Network (R1 to R4) → 4 Ip

Network ID - 12.0.1.144/30

F. Ip - 12.0.1.144/30

L. Ip - 12.0.1.146/30

B. Ip - 12.0.1.147/30

Increment Val = 2

Network (R2 to R4) → 4 Ip

Network ID - 12.0.1.148/30

F. Ip - 12.0.1.149/30

L. Ip - 12.0.1.150/30

B. Ip - 12.0.1.151/30

Increment V = 4 oct

Network (R3 to R4) → 4 Ip

Network ID - 12.0.1.152/30

F. Ip - 12.0.1.153/30

L. Ip - 12.0.1.154/30

B. Ip - 12.0.1.155/30

Next network ID - 12.0.1.156.

NETWORK THEME (129.70.0.0/16).

2nd Floor 200 PC (200 Ip).

Network ID - 129.70.0.0/24

F. Ip - 129.70.0.1/24

L. Ip - 129.70.0.254/24

B. Ip - 129.70.0.255/24

Increment Val = 256

No.

1st Floor 50 PC - 50 Ip.

Network ID - 129.70.1.0/26

F. Ip - 129.70.1.1

L. Ip - 129.70.1.62/26

B. Ip - 129.70.1.63/26

Increment Val = 4th octet - 64

Network (R4 to R5) - 4 Ip.

Network ID - 129.70.1.64/30

F. Ip - 129.70.1.65/30

L. Ip - 129.70.1.66/30

B. Ip - 129.70.1.67/30

Increment Val = 4

Network (R4 to R6) - 4 Ip

Network ID - 129.70.1.68/30 (64 + 4)

F. Ip - 129.70.1.69/30

L. Ip - 129.70.1.70/30

B. Ip - 129.70.1.71/30

Next network ID - 129.0.1.72